1 Foundations

1.1 Sets

1.1.1 Basic notions

In mathematics, a set is a collection of different things, called elements or members of the set. A set may also be called a *collection* or *family*, especially when its elements are themselves sets; this may avoid confusion between the set and its members. A set may be specified either by listing its elements or by giving a property that characterizes its elements, such as for the set of the prime numbers or the set of all students in a given class.

If x is an element of a set S, one says that x belongs to S or *is in* S, and one writes $x \in S$. The statement y *is not in* S is written a $y \notin S$. The axiom of extensionality states that two sets are equal if and only if they have the same elements.

There exists a set with no elements, and extensionality implies that there is only one such set. It is called the *empty set* (or null set) and is denoted $\emptyset$ or $\{\}$.

A singleton is a set with exactly one element. If x is this element, the singleton is denoted $\{x\}$. The sets $\{\emptyset\}$ and $\emptyset$ are different, because the former has one element (namely, $\emptyset$) and the latter has no elements at all.

1.1.2 Specifying a set

Extensionality implies that to specify a set, it suffices either to list its elements or to provide a property that characterize the set's elements among the elements of a possibly larger set.

Roster notation

Roster or *enumeration notation* is a notation introduced by Ernst Zermelo in 1908 that specifies a set by listing its elements between braces, separated by commas. For example, one sees that $\{4, 2, 1, 3\}$ and $\{\text{blue}, \text{white}, \text{red}\}$ denote sets and not tuples because of the enclosing braces.

The notations $\{\}$ for the empty set and $\{x\}$ for a singleton are examples of roster notation.

When specifying a set, all that matters is whether each potential element is in the set or not, so a set does not change if elements are repeated or arranged in a different order. For examle,

$$\{1, 2, 3, 4\} = \{4, 2, 1, 3\} = \{4, 2, 4, 3, 1, 3\}.$$

When there is a clear pattern for generating all set elements, one can use an ellipsis to abbreviate the notation; for example $\{1, 2, 3, \ldots, 10\}$ is a shorthand for $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Set-builder notation

Set-builder notation specifies a set as being the set of all elements that satisfy some logical formula. More precisely, if $P(x)$ is a logical formual depending on a variable x, which evalutates to `true` or `false` depending on the value of x, then

$$\{x|P(x)\}$$

denotes the set of all x for which $P(x)$ is `true`. For example, a set F can be specified as follows:

$$F = \{n|n \text{ is an integer, and } 0 \leq n \leq 19\}.$$

In this notation, the vertical bar $|$ is read as *such that*, and the whole formula can be read as *F is the set of all n such that n is an integer in the range from 0 to 19 inclusive.*

Some logical formulas, such as *S is a set* or *S is set and $S \notin S$* cannot be used in set-builder notation because there is not set for which the elements are characterized by the formula. There are several ways for avoiding the problem. One may prove that the formula defines a set; this is often almost immediate, but may be very difficult.

1.2 Set relations and operations

There are two types of things one can do with sets:

- One can compare two sets and see their relationship. These are called *set relations*. Equality is the first such relation encountered this far.
- One can produce new sets from other sets, usually two of them. These are called *set operations*.

1.2.1 Subsets

As seen before, two sets A and B are equal if and only if they have the same elements, denoted as $A = B$. Apparently, for evere set B, $B = B$.

A *subset* of a set B is a set A such that every element of A is also an element of B. The following are different ways of expressing the same thing:

- A is a subset of B or B is a superset of A,
- $A \subseteq B$ or $B \supseteq A$,
- A is contained in B or B contains A,
- $\forall x(x \in A \Longrightarrow x \in B)$.

The relationship between sets established by $\subseteq$ is called inclusion or containment. Apparently, very set B is a subset of itself, namely $B \subseteq B$.

A set A is a *proper subset* of a set B if $A \subseteq B$ and $A \neq B$; to denote this one writes $A \subset B$. Likewise, one may write $B \supset A$.

Examples

- The set of all humans is a proper subset of the set of all mammals.
- $\{1,3\} \subset \{1,2,3,4\}$
- $\{1,2,3,4\} \subseteq \{1,2,3,4\}$

Properties of containment

- Two sets are equal if and only if they contain each other: $A = B$ is equivalent to ($A \subseteq B$ and $B \subseteq A$).
- The empy set is a subset of every set: $\forall A$, $\emptyset \subseteq A$. In particular: $\forall A \neq \emptyset$, $\emptyset \subset A$.

1.2.2 Basic operations

There are several standard operations that produce new sets from given sets, analogously to how addition and multiplication produce new numbers from given numbers. The operations that are considered in this section are those such that all elements of the produced sets belong to a previously defined set. These operations are commonly illustrated with Euler diagrams and Venn diagrams.

In what follows, a set of sets $\mathcal{S}$ will be used. For example, for sets X, Y, Z, $\mathcal{S} = \{X, Y, Z\}$ is such a set of sets. Care is needed as $X \in \mathcal{S}$ etc.

Intersection

The *intersection* of two sets A and B is set denoted $A \cap B$ whose elements are those elements that belong to both A and B. That is,

$$A \cap B = \{x | x \in A \wedge x \in B\},$$

where $\wedge$ denotes the logical `and`. Apparently, $A \cap B \subseteq A$ and $A \cap B \subseteq B$.

Intersection is associative and commutative; this means that for proceeding a sequence of intersections, one may proceed in any order, without the need of parentheses for specifying the order of operations, for example, $\forall A, B, C$ sets:

$$(A \cap B) \cap C = A \cap (B \cap C) = A \cap B \cap C = \{x | x \in A \wedge x \in B \wedge x \in C\}$$

If $\mathcal{S}$ is set of sets, its intersection, denoted

$$\bigcap \mathcal{S} = \bigcap_{A \in \mathcal{S}} A,$$

is the set whose elements are those elements that belond to all sets in $\mathcal{S}$. That is,

$$\bigcap \mathcal{S} = \left\{ x \middle| \bigwedge_{A \in \mathcal{S}} (x \in A) \right\} = \{x | (\forall A \in \mathcal{S}) x \in A\}.$$

Example: If $\mathcal{S} = \{X, Y, Z\}$, then

$$\bigcap \mathcal{S} = X \cap Y \cap Z.$$

Union

The *union* of two sets A and B is a set denoted $A \cup B$ whose elements are those elements that belong to A or B or both. That is,

$$A \cup B = \{x | x \in A \vee x \in B\},$$

where $\vee$ denotes the logical `or`. Apparently, $A \subseteq A \cup B$ and $B \subseteq A \cup B$.

Union is associative and commutative.

If $\mathcal{S}$ is a set of sets, its union, denoted

$$\bigcup \mathcal{S} = \bigcup_{A \in \mathcal{S}} A,$$

is the set whose elements are those elementes that belong to at least one set in $\mathcal{S}$. That is,

$$\bigcup \mathcal{S} = \left\{ x \middle| \bigvee_{A \in \mathcal{S}} (x \in A) \right\} = \{x | (\exists A \in \mathcal{S}) x \in A\}.$$

Example: If $\mathcal{S} = \{X, Y, Z\}$, then

$$\bigcup \mathcal{S} = X \cup Y \cup Z.$$

Set difference

The *set difference* of two sets A and B, is a set, denoted $A \setminus B$ or $A - B$, whose elements are those elements that belong to A, but not to B. That is,

$$A \setminus B = \{x | x \in A \wedge x \notin B\}.$$

Note that $A \setminus B = A \setminus (A \cap B)$.

When $B \subseteq A$ the difference $A \setminus B = \neg_A B$ is also called the *complement* of B in A. Naively speaking,

$$\neg B = \{x | x \notin B\},$$

but this does not lead to a set as is. When all sets that are considered are subsets of a fixed *universal set* U, the complement $\neg_U A$ is often called the absolute complement of A, and is denoted simply as $\neg A$.

The *symmetric difference* of two sets A and B, denoted $A \triangle B$, is the set of those elements that belong to A or B but not on both:

$$A \triangle B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B).$$

Comprehensive practical exercises

A training team is auditing the prerequisites completed by twelve learners in U. Let $\mathcal{F} = \{P, L, S\}$, record completion of `python` (P), linear algebra (L), and statistics (S), respectively. Let

$$U = \{1, 2, 3, \ldots, 12\},$$
$$P = \{1, 2, 4, 5, 7, 9, 11\},$$
$$L = \{2, 3, 5, 6, 9, 10\},$$
$$S = \{1, 3, 5, 7, 8, 10, 12\},$$
$$R = \{10, 1, 5, 3, 12, 8, 3, 7, 5\}.$$

All sets are considered relative to U unless stated otherwise. Show your reasoning whenever a justification is requested.

Exercise 1. Representations and membership (20 points).

1. Rewrite R without repetitions and in increasing order. Decide whether $R = S$ and justify the decision using extensionality.
2. Expand U without an ellipsis. List $E = \{x \in U \mid x \text{ is even}\}$; describe $H = \{2, 3, 5, 7, 11\}$ in set-builder notation.
3. Mark each statement true or false and justify briefly: $5 \in P$, $\{5\} \in P$, $\{5\} \subseteq P$, $\emptyset \in P$, $\emptyset \subseteq P$, $P \in \mathcal{F}$, $5 \in \mathcal{F}$, and $\{P, L\} \subset \mathcal{F}$.
4. Explain why $\emptyset$ and $\{\emptyset\}$ are different sets.

Solution 1. _

Exercise 2. Relations between sets (25 points).

1. Classify each statement as true or false: $P \cap L \subset P$, $P \subseteq P$, $P \subset P$, and $S \subseteq P \cup L$. For every false statement, provide a counterexample or the corrected relation.
2. For each pair among P, L, and S, decide whether either set contains the other; give a counterexample for each non-containment claim.
3. Verify both $R \subseteq S$ and $S \subseteq R$, and state what follows.
4. Rewrite the true relations $P \cap L \subset P$ and $\{P, L\} \subset \mathcal{F}$ using superset notation. Explain why $\emptyset \subseteq A$ for every set A.

Solution 2. _

Exercise 3. Operations on sets (35 points).

1. Compute in roster notation: $P \cap L$, $P \cup L$, $P \setminus L$, $L \setminus P$, and $P \triangle L$.
2. Compute the family operations $\bigcap \mathcal{F}$ and $\bigcup \mathcal{F}$. Name which result is a singleton and decide whether either result is empty.
3. Use rosters to verify commutativity for $P \cap L$ and $P \cup L$ and verify associativity by comparing both sides of
$$(P \cap L) \cap S = P \cap (L \cap S) \quad \text{and} \quad (P \cup L) \cup S = P \cup (L \cup S).$$
4. Verify from the computed rosters that $P \setminus L = P \setminus (P \cap L)$, and that both $(P \setminus L) \cup (L \setminus P)$ and $(P \cup L) \setminus (P \cap L)$ produce $P \triangle L$.
5. Compute the complement of P in U, written $\neg_U P = U \setminus P$. Explain why the unrestricted expression $\{x \mid x \notin P\}$ does not specify a set unless a containing set is supplied.

Solution 3. _

Exercise 4. Decision rules (20 points). Translate each rule into a set expression or set-builder formula, then give its roster.

1. Learners who completed Python and linear algebra but not statistics.

2. Learners who completed exactly one of Python and linear algebra.
3. Learners who completed all three prerequisites.
4. At least one prerequisite; none. How are these two sets related in U.

Solution 4. _

1.3 Functions

A *function* f from a set A to a set B is a rule that assigns to each element of A a unique element of B.

The notation $f : A \to B$ denotes a function f from A to B. The result of applying f to an element $a \in A$ is denoted $f(a)$; it is called the *value* of f at a, or the *image* of a under f. The set A is called the *domain* of f, and B is called the *codomain* of f.

The graph of a function $f : A \to B$ is the set of all ordered pairs $(a, f(a))$ as a ranged over all elements of A. It is a subset of the Cartesian product $A \times B$.

Indexed family

Intuitively, an indexed family is a set whose elements are labelled with the elements of another (index) set $I \subseteq \mathbb{N}$. These labels allow the same element to occure several times in the family.

An indexed family is a function that has an index set I as its domain. Generally, the usual functional notation is not used for indexed families. Instead, the element i of the index set I is written as a subscript of the same name of the family, such as in a_i. For

- $I = \{0, 1\}$, we have an ordered pair.
- $I = \{0, ..., n-1\}$ with $n \in \mathbb{N}$, we have an n-tuple.
- $I = \mathbb{N}$, we have a sequence.

The above notations

$$\bigcup_{A \in \mathcal{S}} A \text{ and } \bigcap_{A \in \mathcal{S}}$$

are commonly replaced with a notation involving indexed families, namely

$$\bigcup_{i \in I} A_i = \{x | (\exists i \in I\} x \in A_i) \text{ and } \bigcap_{i \in I} A_i = \{x | (\forall i \in I\} x \in A_i).$$

1.3.1 External operations

In section §1.2.2, all elements of sets produced by set operations belong to previously defined sets. In this section, other set operations are considered, which produce sets whose elements can be outside all previously considered sets. These include:

- 1.3.1
- disjoint union
- set exponentiation
- power set

Cartesian product

Given sets A and B, their *Cartesian product* (or simply *product*), denoted $A \times B$, is the set of all ordered pairs (a, b) such that $a \in A$ and $b \in B$; that is,

$$A \times B = \{(a, b) | a \in A \land b \in B\}.$$

The definition makes sense even if $A = B$.

Given an (ordered) indexed family of sets $(A_i)_{i \in I}$, the product

$$\prod\nolimits_{i \in I} A_i$$

is the set of all indexed families of elements $(a_i)_{i \in I}$ such that $a_i \in A_i$, for each $i \in I$.

Set exponentiation

Given two sets A and B, the *set exponentiation*, denoted B^A, is the set that has as elements all functions $A \to B$, i.e.:

$$B^A = \{f | f : A \to B\}.$$

Equivalently, B^A can be viewed as the Cartesian product of a family, indexed by A, of sets that are all equal to F. This explains the terminology and the notation, since exponentiation with integer exponents is a product where all factors are equal to the base:

$$B^A = \prod\nolimits_{a \in A} B = \{(b_a)_{a \in A} | (\forall a \in A) b_a \in B\}.$$

Comprehensive practical exercises

Exercise 5. Functions and graphs (30 points).

A training lab assigns each model run to a compute device. Let the run set, device set, and three candidate assignment graphs be

$$\begin{aligned} R &= \{0, 1, 2\}, \\ D &= \{\text{cpu}, \text{gpu}\}, \\ F &= \{(0, \text{cpu}), (1, \text{gpu}), (2, \text{cpu})\}, \\ G &= \{(0, \text{cpu}), (1, \text{gpu})\}, \\ H &= \{(0, \text{cpu}), (0, \text{gpu}), (1, \text{cpu}), (2, \text{gpu})\}. \end{aligned}$$

1. Determine which of F, G, and H is the graph of a function from R to D. For each invalid candidate, identify whether existence or uniqueness fails and where.
2. Let f be the function represented by the valid graph. Write its declaration, state its domain and codomain, compute $f(0)$, $f(1)$, and $f(2)$, and name the image of 1 under f.
3. Construct $R \times D$ in roster notation and verify directly that the graph of f is a proper subset of it.

Solution 5. _

Exercise 6. Indexed families (30 points).

Consider an indexed family of status labels and an indexed family of sets:

$$\begin{aligned} I &= \{0, 1, 2, 3\}, \\ C &= \{\text{red}, \text{blue}, \text{green}\}, \\ (a_i)_{i \in I} &= (\text{red}, \text{blue}, \text{red}, \text{green}), \\ J &= \{0, 1, 2\}, \\ A_0 &= \{1, 2, 4\}, \\ A_1 &= \{2, 3, 4\}, \\ A_2 &= \{2, 4, 5\}. \end{aligned}$$

1. View $(a_i)_{i \in I}$ as a function $a : I \to C$ and list its four values. Explain why red may occur twice in the indexed family even though a set does not contain repeated elements.
2. Classify $(a_i)_{i \in I}$ as an ordered pair, an n-tuple, or a sequence. State what the corresponding family would be called if its index set were instead $\{0, 1\}$ or $\mathbb{N}$.
3. Compute $\bigcup_{i \in J} A_i$ and $\bigcap_{i \in J} A_i$ in roster notation. Also write each operation as a set-builder condition using the appropriate existential or universal quantifier.
4. Use the quantifier meanings to explain why 3 belongs only to the indexed union, while 2 belongs to both.

Solution 6. _

Exercise 7. Cartesian products and set exponentiation (40 points).

Let

$$\begin{aligned} X &= \{a, b\}, \\ Y &= \{0, 1\}, \\ K &= \{0, 1, 2\}, \\ (C_i)_{i \in K} &= (\{a, b\}, \{0, 1\}, \{*\}). \end{aligned}$$

1. Compute $X \times Y$, $Y \times X$, and $X \times X$ in roster notation. Decide whether the first two products are equal and explain the role of coordinate order.
2. Compute the indexed product $\prod_{i \in K} C_i$ by listing every indexed triple (c_0, c_1, c_2).
3. List all elements of Y^X by representing each function $X \to Y$ first by its graph and then by the indexed pair $(f(a), f(b))$.
4. Verify concretely that $Y^X = \prod_{x \in X} Y$. Explain which set supplies the domain, which supplies the codomain, and why this example contains four functions.
5. Why are Cartesian product and set exponentiation external operations?

Solution 7. _